

ASEN 6044

Homework #2

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Problem 1:

Problem 2:

Problem 3:

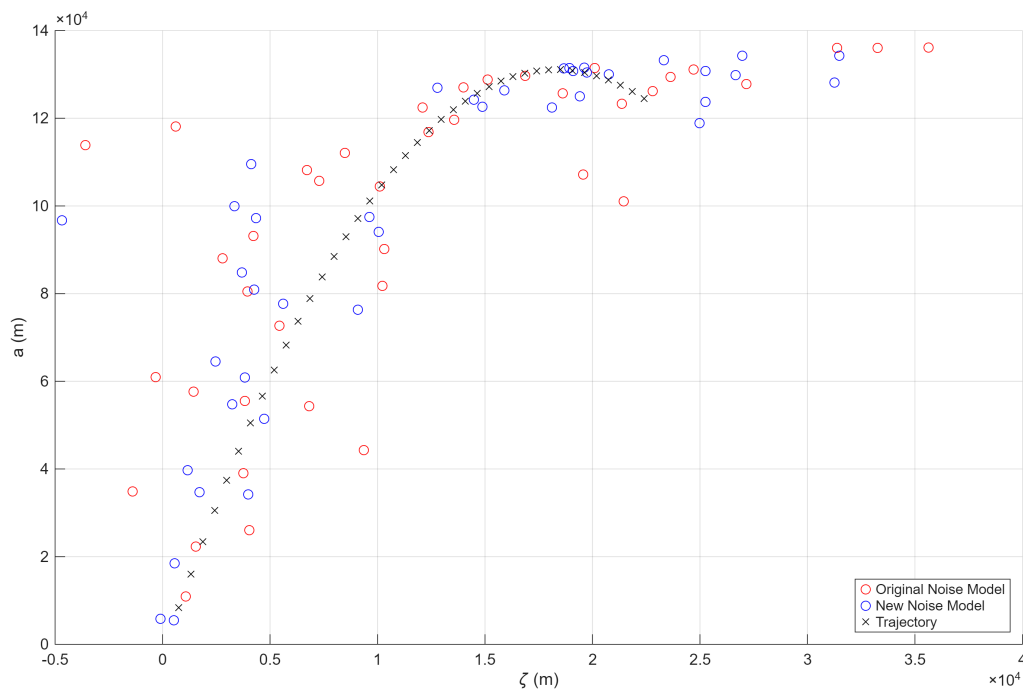


Figure 1: Comparison of Old and New Noisy Measurements

- From Figure 1 we see generally that the measurements from the new noisy range model draw out a similar shape to that of the old range model. Without any prior knowledge regarding the problem, it even appears that both sets of measurements have the same noise model.
- TODO: Best x_0 , Variance of range? Is it even supposed to work??
- The negative log-likelihood function, $\mathcal{L}(x_0; Y_{1:N}) = J_{NLL}$ is defined as the negative logarithm of the measurement likelihood functions, that is

$$J_{NLL} = -\log(P(Y_{1:N} | x_0))$$

To derive an expression for $P(Y_{1:N} | x_0)$, we look at the given measurement model:

$$\rho_k = \sqrt{(L - \xi(t_k))^2 + (a(t_k))^2} + v_{r,k} \quad v_{r,k} \sim \mathcal{T}_{v_{r,k}}\left(0, \nu = \frac{1}{2}\right) \quad (1)$$

$$\theta_k = \arctan\left(\frac{a(t_k)}{L - \xi(t_k)}\right) + v_{b,k} \quad v_{b,k} \sim \mathcal{N}_{v_{b,k}}\left(0, \sigma_b^2 = 0.005\right) \quad (2)$$

Where $v_{r,k}$ and $v_{b,k}$ are the range and bearing noise respectively. To derive the measurment likelihood $P(Y_{1:N} | x_0)$ we assume that the measurments are conditionally independent given x_0 , that is $P(Y_{1:N} | x_0) = \prod_{k=1}^N P(y_k | x_0)$. We also assume that the range and bearing measurments are independent of each other, our expression further simplifies to the following:

$$P(Y_{1:N} | x_0) = \prod_{k=1}^N P(\rho_k | x_0) P(\theta_k | x_0) \quad (3)$$

Since our noise is additive, solving for the respective probabilities is fairly straightforward. From (1) we can derive $P(\rho_k | x_0)$ as follows:

$$v_{r,k} = \rho_k - \sqrt{(L - \xi(t_k))^2 + (a(t_k))^2}$$

$$P(v_{r,k}) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}} \left(1 + \frac{v_{r,k}^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

Substituting for $v_{r,k}$ we get the following expression in terms of our initial conditions.

$$P(\rho_k | x_0) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}} \left(1 + \frac{(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 t_k - 0.5gt_k^2)})^2}{\nu}\right)^{-\frac{\nu+1}{2}} \quad (4)$$

We can follow a similar process to derive $P(\theta_k | x_0)$.

$$v_{b,k} = \theta_k - \arctan\left(\frac{a(t_k)}{\xi(t_k)}\right)$$

$$P(v_{b,k}) = \frac{1}{\sigma_b\sqrt{2\pi}} \exp\left\{\frac{-1}{2\sigma_b^2} v_{b,k}^2\right\}$$

$$P(\theta_k | x_0) = \frac{1}{\sigma_b\sqrt{2\pi}} \exp\left\{\frac{-\left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)\right]^2}{2\sigma_b^2}\right\} \quad (5)$$

Substituting (5) and (4) into (3) and taking the negative logarithm we recieve the following

expression for our negative log likelihood function:

$$\begin{aligned}
 \mathcal{L}(x_0; Y_{1:N}) &= - \sum_{k=1}^N \log[P(\rho_k | x_0)P(\theta_k | x_0)] \\
 &= - \sum_{k=1}^N \log\left(\frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}} \frac{1}{\sigma_b\sqrt{2\pi}}\right) \\
 &\quad - \sum_{k=1}^N -\frac{\nu+1}{2} \log\left(1 + \frac{(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 t_k - 0.5gt_k^2)})^2}{\nu}\right) \\
 &\quad - \sum_{k=1}^N \frac{\left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)\right]^2}{2\sigma_b^2}
 \end{aligned} \tag{6}$$

Both the NLL and NLS cost functions seek to find an x_0 that minimizes the cost function given observations $Y_{1:N}$, however they each approach the problem in a different way. NLS is defined to minimize the sum of the squared residuals, while maximum likelihood tries to find the x_0 which makes ‘the most sense’ given all the observations. Both assume the measurement noise characteristics are known, NLS uses a least squares cost function, meaning we only consider the 1st/2nd order moments. As we have seen with the Student’s-t distribution, this does not accurately capture higher order moments. Maximum likelihood is robust to non-Gaussian sensor noise unlike the least-squares cost function we have selected for NLS.

d) In order to minimize (6), we take the gradient with respect to x_0 , or $\nabla_{x_0}\mathcal{L}(x_0; Y_{1:N})$, where

$$\nabla_{x_0} = \begin{bmatrix} \frac{\partial}{\partial \xi_0} \\ \frac{\partial}{\partial \dot{\xi}_0} \\ \frac{\partial}{\partial a_0} \\ \frac{\partial}{\partial \dot{a}_0} \end{bmatrix}$$

$$\nabla_{x_0}\mathcal{L}(x_0; Y_{1:N}) = -\nabla_{x_0} \sum_{k=1}^N \log[P(\rho_k | x_0)P(\theta_k | x_0)]$$

It’s easy to show that the expression simplifies to the following, where the constant terms evaluate to zero.

$$\begin{aligned}
 &= -\nabla_{x_0} \left[\frac{-\nu+1}{2} \log\left(1 + \frac{(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 t_k - 0.5gt_k^2)})^2}{\nu}\right) \right] \\
 &\quad - \nabla_{x_0} \left[\frac{-1}{2} \left[\frac{\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)}{\sigma_b} \right]^2 \right]
 \end{aligned} \tag{7}$$

Equation (7) consists of two gradient terms to evaluate. Starting with the first term we can take each partial derivative independently and stack the result into a vector:

$$\begin{aligned}
& \nabla_{x_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\
& \frac{\partial}{\partial \xi_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\
& = \frac{2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \\
& \frac{\partial}{\partial \dot{\xi}_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\
& = \frac{2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \\
& \frac{\partial}{\partial a_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\
& = \frac{-2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \\
& \frac{\partial}{\partial \dot{a}_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\
& = \frac{-2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}}
\end{aligned}$$

Therefore,

$$\nabla_{x_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right]$$

$$= \frac{-\nu + 1}{2} \left[\frac{2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}} \right. \\ \left. \frac{2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}} \right. \\ \left. \frac{-2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}} \right. \\ \left. \frac{-2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}} \right] \quad (8)$$

We will follow a similar process for the second gradient term.

$$\nabla_{x_0} \left[\frac{-1}{2} \left[\frac{\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)}{\sigma_b} \right]^2 \right] \\ = \frac{-1}{2\sigma_b^2} \nabla_{x_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 \\ \frac{\partial}{\partial \xi_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 \\ = \frac{-2t[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)](a_0 + \dot{a}_0 t_k - 0.5gt_k^2)}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ \frac{\partial}{\partial \dot{\xi}_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 \\ = \frac{-2[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)](a_0 + \dot{a}_0 t_k - 0.5gt_k^2)}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ \frac{\partial}{\partial a_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 \\ = \frac{-2[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)]}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)}$$

$$\begin{aligned}
& \frac{\partial}{\partial \dot{a}_0} \left[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right) \right]^2 \\
&= \frac{-2t[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)]}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)}
\end{aligned}$$

Therefore,

$$\begin{aligned}
& \nabla_{x_0} \left[\frac{-1}{2} \left[\frac{\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)}{\sigma_b} \right]^2 \right] \\
&= \frac{-1}{2\sigma_b^2} \left[\begin{aligned} & \frac{-2[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)](a_0 + \dot{a}_0 t_k - 0.5 g t_k^2)}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ & \frac{-2t[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)](a_0 + \dot{a}_0 t_k - 0.5 g t_k^2)}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)^2} \\ & -2[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)] \\ & \frac{\left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)^2 + 1}{(L - \xi_0 - \dot{\xi}_0 t_k)} \\ & \frac{-2t[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)]}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \dot{\xi}_0 t_k)} \end{aligned} \right] \quad (9)
\end{aligned}$$

Finally, we can substitute equations (8) and (9) into equation (7) and we are left with an expression for the gradient vector of $\mathcal{L}_{NLL}$

$$\begin{aligned}
\nabla_{x_0} \mathcal{L}(x_0; Y_{1:N}) = & \sum_{k=1}^N \left[\frac{\nu - 1}{2} \left[\frac{2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt^2)^2}} \right. \right. \\
& \frac{2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt^2)^2}} \\
& \frac{-2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt_k^2)^2})(a_0 + a_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt^2)^2}} \\
& \left. \frac{-2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt_k^2)^2})(a_0 + a_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0 t)^2 + (a_0 + a_0 t - 0.5gt^2)^2}} \right] \\
& + \frac{1}{2\sigma_b^2} \left[\frac{-2[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k}\right)](a_0 + a_0 t_k - 0.5gt_k^2)}{\left(\left[\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \xi_0 t_k)^2} \right. \\
& \frac{-2t[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k}\right)](a_0 + a_0 t_k - 0.5gt_k^2)}{\left(\left[\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \xi_0 t_k)^2} \\
& \frac{-2[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k}\right)]}{\left(\left[\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \xi_0 t_k)} \\
& \frac{-2t[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k}\right)]}{\left(\left[\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \xi_0 t_k)} \\
& \left. \frac{-2t[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k}\right)]}{\left(\left[\frac{a_0 + a_0 t_k - 0.5gt_k^2}{L - \xi_0 - \xi_0 t_k} \right]^2 + 1 \right) (L - \xi_0 - \xi_0 t_k)} \right]
\end{aligned} \tag{10}$$

e)

f)

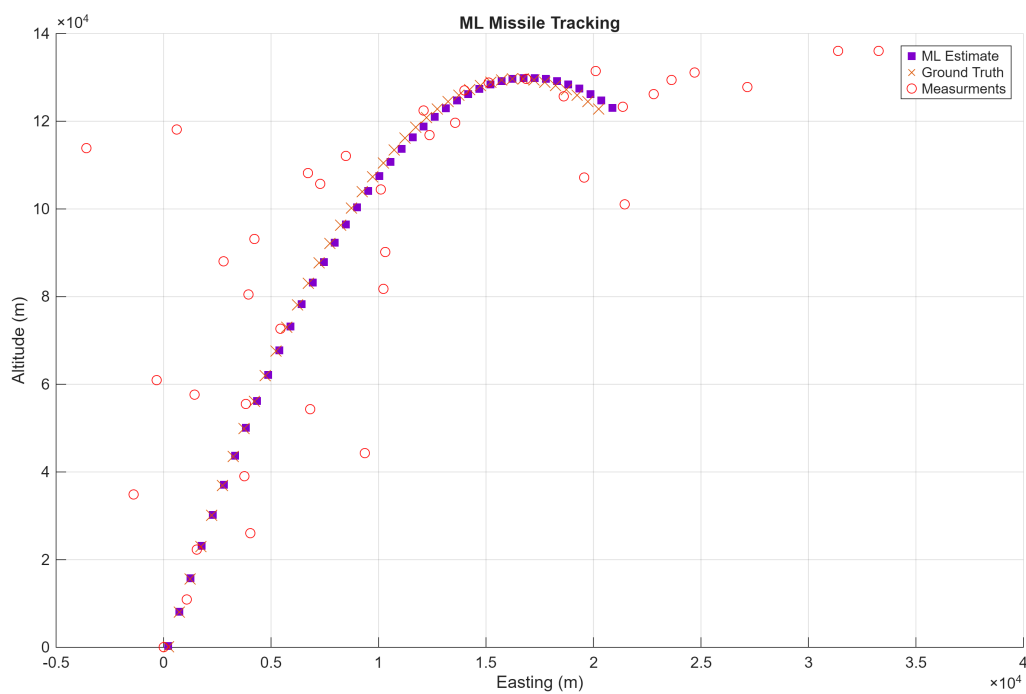


Figure 2: Maximum Likelihood Estimated Trajectory